GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Taal

The agent that sells what the forecast says you'll throw away.

TEAM

Nandish · lead

Rinu · co-builder

PROBLEM

1. Forecasts and offers live in different tools: one dark store writes stock off while the next runs out.
2. Customers get blanket markdowns, not offers matched to what they buy.
3. Nothing plans to the online sell-by cut-off: 30% of shelf life or 45 days at delivery.[1]

[1] FSSAI advisory to e-commerce FBOs, 3 Dec 2024, file RCD-13/1/2024-Regulatory-FSSAI(E-13150): minimum 30% shelf life or 45 days at delivery. Either-limit reading is Taal's configurable default.

BRIEF ABOUT THE IDEA

One decision loop turns a forecast into a sale, before the write-off happens.

1. **Capture** a pallet photo
2. **Sense** nightly gaps, in rupees
3. **Plan** a play; the estimator owns the rupees
4. **Approve** with a holdout; feed it back to the forecast
5. **Engage** consenting customers, in their language
6. **Measure** treated vs holdout, 95% CI

THEME **Taal forecasts stock per SKU and node nightly, with festival effects, and drafts human-approved offers.**

ONE EXAMPLE · SEEDED DEMO TENANT

6 days

until Masala Chips at dark store DS-07 can no longer be sold online

368	₹9,200
units	at stake

Recorded Gemini run, 10 Oct: one schema rejection, then all eight guardrails pass. Approve: 323 treated, 29 holdout.

DIFFERENTIATION

A holdout on every play. The play written back into the forecast.

	ONLINE SELL-BY CUT-OFF	HOLDOUT PER PROMO	PROMO → FORECAST WRITE-BACK	INDIA / Q-COMMERCE
Taal	Yes	Yes	Yes	Built for it; seeded tenant
Blue Yonder	Unclear	Unclear	Unclear	Partial
RELEX	Partial	Unclear	Partial	Partial
Wasteless	No	Unclear	Unclear	Unclear
Afresh	No	Unclear	Unclear	No
Flashfood	No	No	No	No

Unclear = no public evidence found in vendor pages read on 2026-10-09; it does not mean the feature is absent. Partial and No are our reading of those pages. Afresh and Flashfood are different product categories.

LIST OF FEATURES

Six capabilities, gated by rules a model never touches.

Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.

Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.

Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.

Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.

Customer Agent

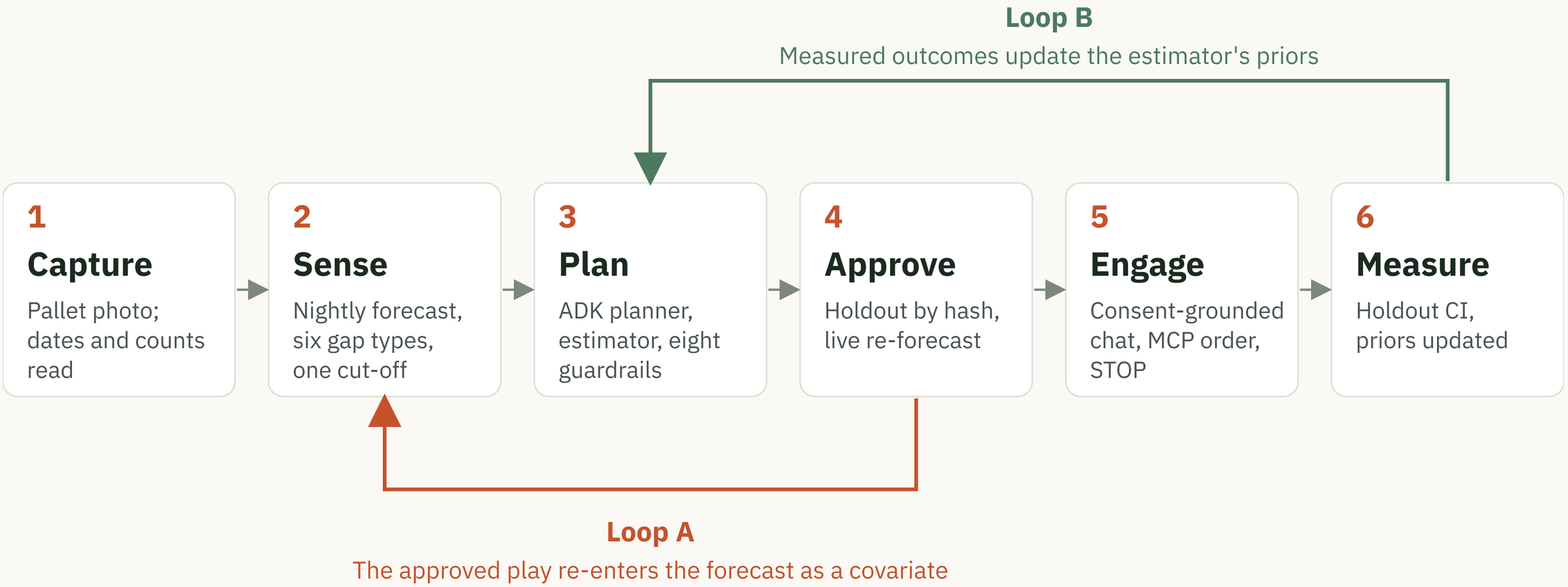
Grounded in stock, offers and consent. Orders through an in-process MCP tool.

Measure

Treated vs holdout, 95% CI. Priors update from the counts.

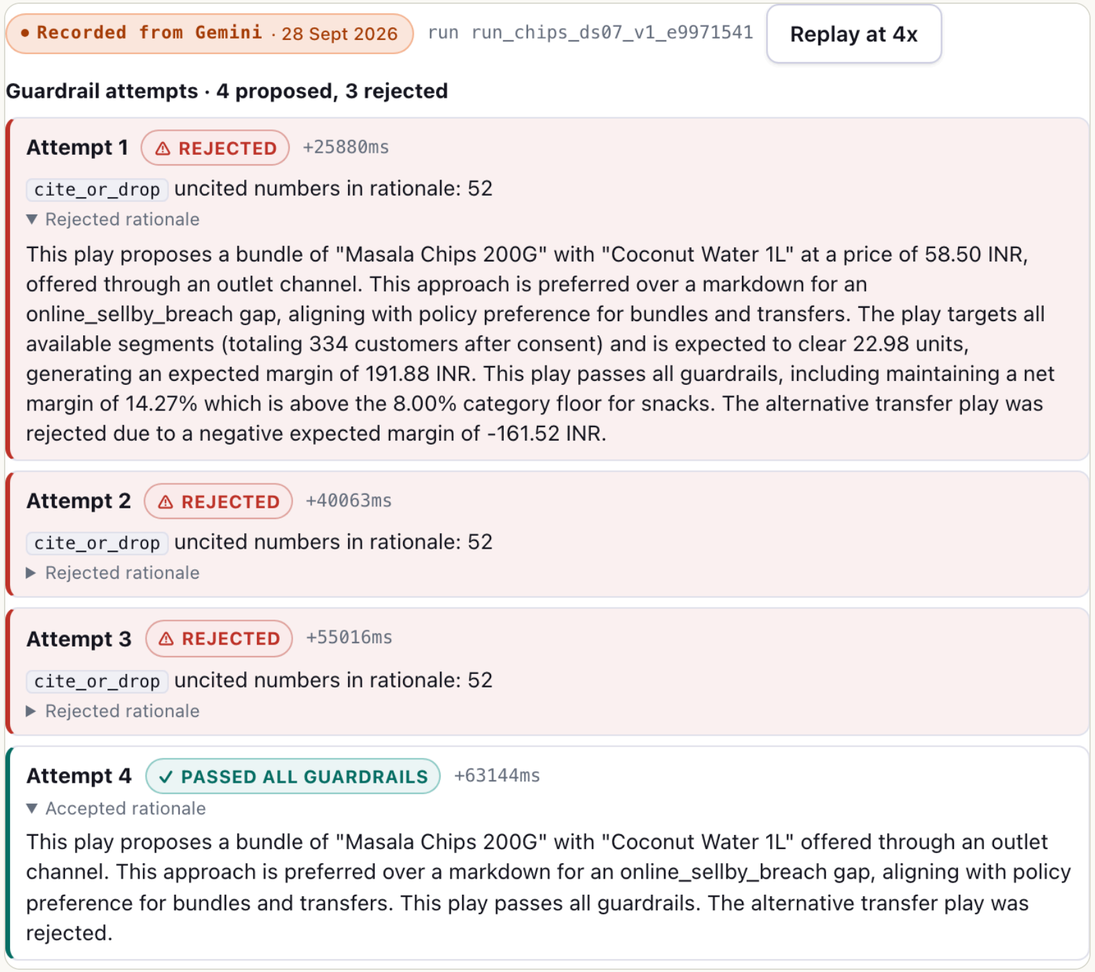
PROCESS FLOW

Two closed loops, not an open-ended pipeline.



SCREENS

Guardrails reject; customers get Kannada plus English.





Right: customer chat, live

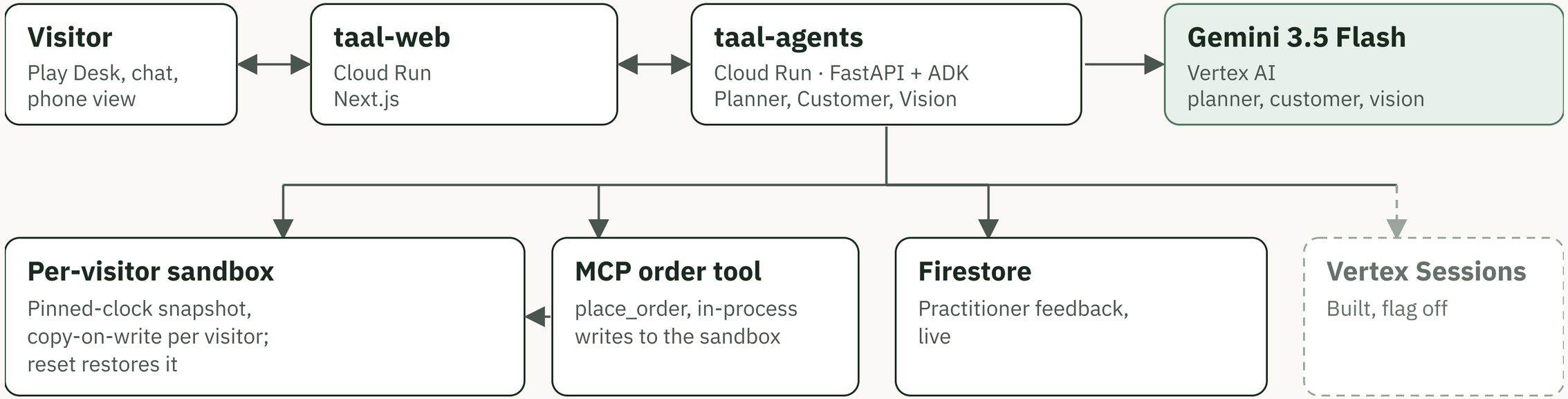
Kannada reply with an English gloss; the order goes through the MCP order tool.

Left: Play Desk trace, recorded run (28 Sep)

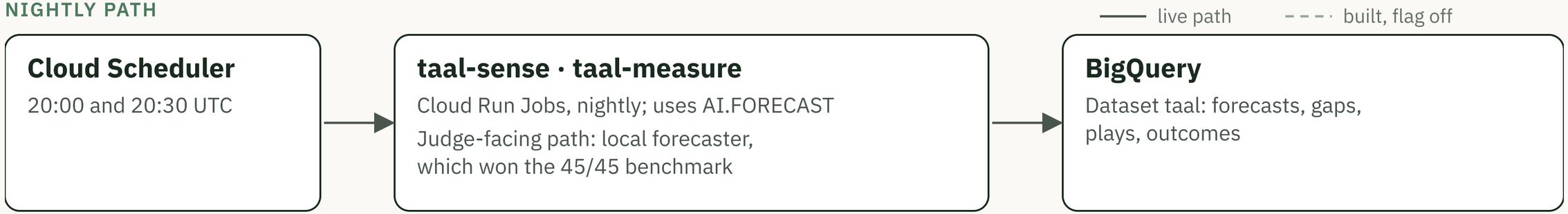
28 Sep run, before the tokenizer fix: three attempts rejected by cite-or-drop, the fourth passes. The 10 Oct run needed one schema retry.

ARCHITECTURE

A sandbox serves judges; nightly jobs run on BigQuery.



NIGHTLY PATH



TECHNOLOGIES USED

Google Cloud AI stack, benchmarked before use.

Gemini 3.5 Flash

On Vertex AI: vision intake, Planner Agent and Customer Agent.

Google ADK

LoopAgent (max 3), 6 deterministic tools and 8 guardrails around every play.

BigQuery

AI.FORECAST benchmarked and run nightly; AI.GENERATE_TABLE writes offer copy. Judges see the local forecaster.

Firestore

Practitioner feedback store, live in the deployed service.

Cloud Run

Services scale to zero; Cloud Run Jobs and Cloud Scheduler run Sense and Measure nightly.

Next.js · FastAPI · MCP

Play Desk, phone view and chat on Next.js; API on FastAPI; orders via an in-process MCP tool.

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

Priced from measured usage; sold on rupees rescued.

PER 30 DAYS, MODELLED FROM MEASURED USAGE

₹486 / $5.86

Real Gemini tokens, 752 BigQuery jobs and Cloud Run time, priced at public list rates. Not a bill.

Method: modeled_from_measured_usage.

Forward-looking profiles ESTIMATES

Profile	Scale	Monthly
Micro	1 shop, ≤500 SKUs	₹200–500
Mid	300 SKUs, 10 nodes	₹2,500–5,000
Large	20,000 SKUs, 500 nodes	₹1–2 lakh

BUYER AND PRICING

Buyer: the category or supply-chain head, who owns write-off.
Basis: a fee per node per month, or a percentage of rupees rescued vs the holdout.

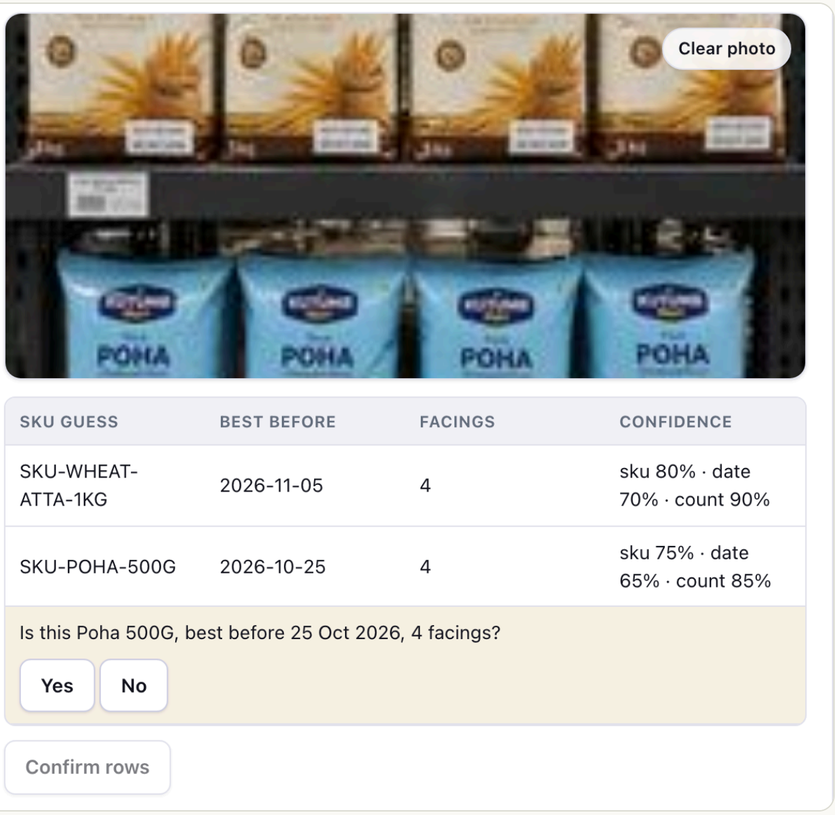
BREAK-EVEN ILLUSTRATION

≈ ₹21,710 – ₹30 per node

Per month: projected margin ₹347,356 and run cost ₹486, each over 16 nodes. Synthetic, not a quote.

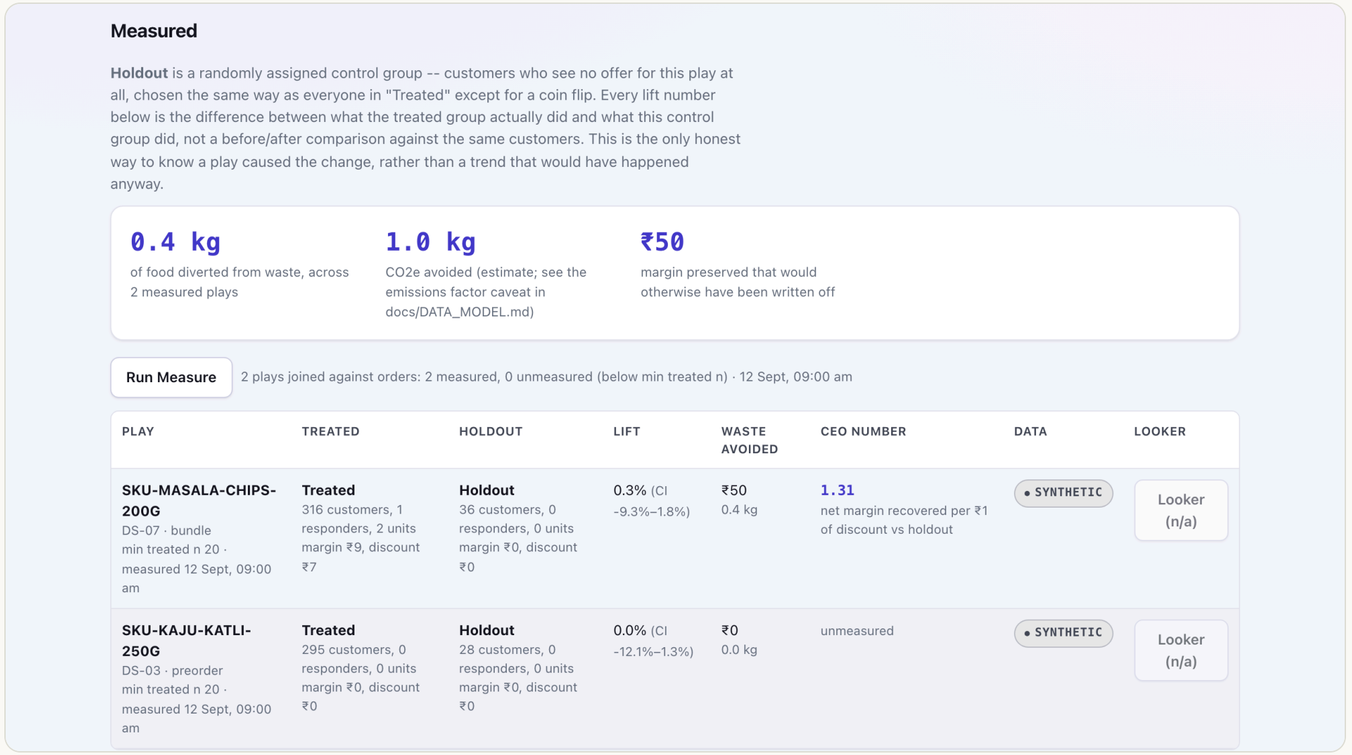
SNAPSHOTS OF THE PROTOTYPE

Doubtful reads go to a person; outcomes say SYNTHETIC.



Phone view

A read under 70% confidence is asked back before it reaches inventory.



Outcomes: treated vs holdout, SYNTHETIC badge

Screenshot from the 28 Sep build (316 / 36). The current seeded example is 323 treated, 29 holdout; the lift interval crosses zero and is reported as such.

PERFORMANCE REPORT

Measured, dated, and traced to a raw file.

PLANNER, LIVE — 10 Oct

5/5

proposed, 0 fallbacks; 22–28 s each

`eval/raw/planner_real_traces_2026-10-10/summary.json`

CUSTOMER AGENT — 28 Sep

4.56 s p50

p95 6.48 s, 0.48 s over the 6 s budget; 50/50 returned 200

`eval/raw/customer_latency_2026-09-28/analysis.json`

APPROVE + RE-FORECAST — 21 and 28 Sep

8.3–10.6 s

two runs; write-off ₹9,194 → ₹8,068

`eval/raw/sweep_vertex_2026-09-21.txt` · `eval/raw/post_deploy_check_2026-09-28/summary.json`

AGENT SIMULATION — 21 Sep

190/190

live Vertex; stock, holdout arm, stacking, STOP, hallucination

`eval/raw/agent_simulation_2026-09-21.json`

VISION INTAKE — 21 Sep

30/30

correct dates; synthetic photos only

`eval/raw/vision_synthetic_2026-09-21.json`

BACKTEST, SYNTHETIC — 20 Sep

0.113 WAPE

MAPE 0.133 over 63 tier-origin rows

`eval/raw/backtest_rows_2026-09-20.jsonl` · `eval/raw/external_backtest_2026-10-09.json`

PUBLIC NON-GROCERY DATASET BACKTEST — UCI Online Retail II

0.83 WAPE

Method and caveats in eval/external_backtest.md

`eval/raw/external_backtest_2026-10-09.json`

LIVE RE-PLAN, DEPLOYED — 10 Oct

30.3 / 27.5 / 29.6 s

3 live runs, 0 fallbacks, 1 loop iteration each; 90 s deadline

47.7 / 47.7 / 30.8 s

before the fix, 9 Oct: 2 of 3 hit a 45 s deadline and fell back to rules

`eval/raw/live_replan_timing_2026-10-10.json` · `eval/raw/live_replan_timing_2026-10-09.json`

LIVE EVAL ON VERTEX — 2026-10-09

15/15

rationale judge; same-family judge, weak evidence

38/46

ADK response match

6/46

exact trajectory; retry-order variance

`eval/raw/live_eval_2026-10-09/summary.json`

VALIDATION · PRACTITIONER RESPONSES, SEPTEMBER 2026

What seven practitioners told us.

3 of 7
write off stock weekly or daily

3 of 6
online sellers say the sell-by rule limits what they sell

0 of 7
measure a promotion against a control group

4 of 7
want a one-week pilot

"Blanket markdowns erode margin on products that may still sell normally."

Category / merchandising, quick-commerce

"Item-store recommendations with incremental-margin estimates."

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

USER GUIDE

Three people, three screens, one approved play.

Planner

/desk

1. Open the Play Desk and pick a gap.
2. Check the guardrails and counterfactuals.
3. Click Approve. The forecast moves.

Node manager

/phone

1. Photograph a pallet, or tap a sample.
2. Confirm the rows read below 0.7 confidence.
3. Approve the play from the gap card.

Customer

/chat

1. Open the chat. It starts with "Any offers today?".
2. Ask about a product. You see a stock band, not a count.
3. Send STOP to withdraw consent.

Badges. LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

IMPACT · WHAT IS MEASURED, WHAT IS PROJECTED

Projected margin hinges on one assumption.

TWO LOTS, 10 OCT

Tea (the 60-second beat)

+₹4,493

103 units, ₹35,020 at stake. Play net −₹27,522 against −₹32,015 for a blanket 20% markdown.

Chips (recorded Gemini run)

-₹279

368 units, ₹9,200 at stake. Play net −₹8,468 against −₹8,189. The Desk shows the loss.

RESPONSE PRIOR SENSITIVITY · PROJECTED

PRIOR	MARGIN	WASTE AVOIDED
2%	₹1.41 lakh	₹3.26 lakh
5% default	**₹3.47 lakh**	**₹7.06 lakh**
10%	₹6.19 lakh	₹11.18 lakh

Seeded tenant, 28-day horizon, 550 grocery gaps; 421, 424 and 423 plays planned.

MEASURED VS PROJECTED

Measured on real customers: none yet.

Synthetic holdout: 323 treated, 29 held back; lift +1.5 points, 95% CI −12.4 to +5.6, crosses zero.

Projected at 5%: ₹347,356 margin, ₹706,183 waste avoided.

Practitioners: 7 usable responses; 3 of 7 write off stock weekly or daily.

LONGEVITY · RUNNING IT FOR REAL

What it takes to run past the competition.

Production path

Least-privilege accounts replace the deploy owner.

CI uses a stored key. Move to workload identity.

Secrets sit in a managed store.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

Scalability

Measured: Sense 3.8 s; planner re-plan 27 to 30 s live; chat 4.5 to 6.9 s.

Estimate: 20,000 SKUs, 500 nodes. ESTIMATE

Breaks first: the planner queue.

Feasibility

Buyer: category or supply-chain head. Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Run cost: ₹486 a month, measured usage, not a bill.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

ADDITIONAL DETAILS · FUTURE DEVELOPMENT

Next: a real pilot, real photos, more tenants.

1

A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout. Metric pre-registered in docs/pilot.md.

DESIGNED

2

Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 is on synthetic ones.

HARNESS BUILT

3

Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LINKS

Taal: the agent that sells what the forecast says you'll throw away

GITHUB REPOSITORY



https://github.com/Nandish3010/probable-fortnight

LIVE APP



https://taal-web-2obkp776ca-el.a.run.app

Open the live app, click Run the 60-second beat, then Approve. Nothing you do there is saved.

Team Taal Google Cloud AI Builder Cup 2026 · Retail & Commerce